

Concentration Types and Dilution Activity {LS25}

Around the room are various stations with products or equipment. As you tour around to the stations, complete the following sections.

Station 1: Parts per Million (ppm) $\text{ppm} = (\text{mg}_{\text{solute}} / \text{kg}_{\text{water}})$ OR $\text{ppm} = (\text{g}_{\text{solute}} / \text{g}_{\text{water}}) \times 10^6$
Ratio of mg/kg = 1:10⁶

- 1) List the PPM of two chemicals (ions) in the water bottles
- 2) Using the volume of the bottle (convert to mass), find the mass of the ions that would be found in a filled bottle.
- 3) Using (2), calculate the number of ions (particles) for each that would be found in a filled bottle.

Station 2: Mass/Mass %

$$\text{Equation: } m\% = (m_{\text{solute}} / m_{\text{total}}) \times 100\%$$

- 1) For each product, select 1-2 ingredients. Find their concentration in m/m% in a SERVING using the serving mass and the mass of the ingredient. Mass must be in the same units (ie. g/g, or kg/kg).

Product	Ingredient	Serving Mass	Mass of Ingredient / Serving	Mass / Mass %

Station 3: Mass/Volume %

$$\text{Equation: } m\% = (m_{\text{solute}}(\text{g}) / V_{\text{total}}(\text{mL})) \times 100\%$$

- 2) For each product, select 1-2 ingredients. Find their concentration in m/v in a SERVING using the serving volume (in mL) and the mass of the ingredient (use grams per mL)

Product	Ingredient	Serving Volume	Mass of Ingredient / Serving	(Mass / Volume) x 100%

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Station 4: Volume/Volume %

$$\text{Equation: } v/v\% = (V_{\text{solute}} / V_{\text{total}}) \times 100\%$$

- 3) For each product, select 1 ingredient. Find the amount of the ingredient in the filled container using the volume of the container and the %v/v listed. Volumes must be in the same units (ie. mL/mL or L/L)

Product	Ingredient	Total Volume	% v/v of Ingredient	Volume of Ingredient in Container

Station 5: Dilution (using mass)

Weigh out 0.2g of NaCl. Add to the volumetric flask. Filled the flask half way to the mark shown on the stem of the bottle. Allow the salt to dissolve by swirling the flask. Then fill the rest of the bottle with water to the mark. Calculate the concentration of NaCl in moles/L you have made in the flask. When you are finished, empty the bottle in the sink. Complete the questions below.

- 1) Calculation:
- 2) What was the solute: _____. What is the solvent: _____
- 3) If we were to double the solute, what will happen to the concentration of any solution?
- 4) If we were to double the solvent, what will happen to the concentration of any solution?